

## (a) The Consistency Problem

Shared Premise: "If it rains, Alice goes to the gym."



Q1

"If it rains, does Alice go to the gym?"

Direct implication

"Yes" ✓ CORRECT

Q2

"If Alice did NOT go to the gym,  
can we conclude it didn't rain?"

Contrapositive (logically equiv. to Q1)

"No" ✗ WRONG

Q3

"If Alice went to the gym,  
can we conclude it rained?"

Affirming the consequent (fallacy)

"Yes" ✗ WRONG

Q1 & Q2 are logically equivalent (contrapositive), but the model contradicts itself

Individual Accuracy

$1 / 3 = 33\%$

CONSISTENCY  
GAP

Set Consistency Rate

$0 / 1 = 0\%$

## (b) The Consistency Gap

